

AKILAN M

Aspiring Data Engineer

Bangalore | +91 6383101826 | akilanjai71@gmail.com | LinkedIn - [linkedin.com/in/akilan-m-672b30319](https://www.linkedin.com/in/akilan-m-672b30319) | github.com/akilanm71

SUMMARY:

Aspiring Data Engineer with proficiency in Python, SQL, Machine Learning, and analytical tools, backed by a strong foundation in statistical methods and Data analysis skills. Hands-on experience working with real-world business data during internship at [Rubix](#), gaining practical exposure to data analysis, business insights, and industry-level workflows. Skilled in transforming complex data into actionable insights through data-driven solutions, with a strong passion for continuous learning and building scalable AI and analytics solutions.

WORK EXPERIENCE:

RUBIXE AI Solutions

Data Science Intern | September 2025 – March 2026

- Worked on various proof-of-concept and client projects, applying data science techniques to solve real-world problems.
- Demonstrated strong work ethic by consistently delivering tasks on time and meeting project deadlines.
- Collaborated with cross-functional teams to understand project requirements and develop data driven solutions.

Biocon Biologics Bangalore

Senior Executive Pharmaceutical Operations | August 2023 –June 2025

- Analysed large-scale production and quality datasets using Excel to identify trends and process inefficiencies.
- Converted raw operational data into structured reports and summaries to support data-driven decision-making.
- Monitored process metrics and compliance indicators, ensuring data accuracy and validation across workflows

PROJECTS:

Customer Churn Prediction System (Telecom)

Toolset: Python, Scikit-learn, UMAP, HDBSCAN, LLM, FastAPI, Streamlit

- Performed end-to-end EDA, feature engineering, and churn analysis to identify key factors influencing customer attrition and business retention insights.
- Built and evaluated multiple ML models using ROC and Precision-Recall metrics, achieving **96.3% accuracy, 95.4% precision, and 90.3% recall**.
- Implemented customer segmentation and predictive analytics to compare actual vs predicted churn behaviour for improved decision-making. Developed and deployed an AI-powered churn analytics platform with BERTopic-based complaint analysis, interactive dashboards, and real-time visualizations on Hugging Face spaces

Live App: https://huggingface.co/spaces/akilanjai71/churn_prediction_labeling

Streamlit Dashboard: <https://customerchurndas3hboard-82dgvxx9ezq6grv5kgkfbu.streamlit.app>

ITSM Incident Analytics & Ticket Prediction System

Toolset: Python, Scikit-learn, Time series Analysis ARIMA, Prophet, Deep Learning LSTM, GRU

- Modeled Machine Learning and Time-Series forecasting solutions to improve ITSM operations through predictive analytics, automation, and incident trend forecasting.
- Built predictive models for incident automation and time-based forecasting using XGBoost, Random Forest, ARIMA, Prophet, and LSTM, achieving up to **72.6% accuracy, 88% precision, and 74% recall** for high-priority incident prediction.
- Developed automated preprocessing and feature engineering pipelines by incorporating imbalance handling, cyclic time-feature engineering, and data leakage prevention techniques.

- Optimized forecasting and classification models using GridSearchCV, TimeSeriesSplit, SMOTE, ROC-AUC, and F1-Score evaluation metrics to improve predictive performance and model reliability.

Leaf Disease Classification using Deep Learning and Transfer Learning

Toolset: Image Processing CNN with TensorFlow

- Developed deep learning-based image classification models using CNN and MobileNetV2 to identify and classify plant leaf diseases from image datasets.
- Built efficient image preprocessing and augmentation pipelines using TensorFlow and Keras, incorporating normalization, resizing, batching, shuffling, transfer learning, fine-tuning, and model checkpointing techniques.
- Evaluated custom CNN and pretrained MobileNet models using performance metrics such as accuracy, confusion matrix, and classification reports. Achieved **92% training accuracy** and **99.16 % testing accuracy**, and integrated the best-performing model into a Flask-based deployment application.

House Loan Credit Risk Prediction and Loan Default Analysis

Toolset: Python, SQL, Pandas, NumPy, Scikit-learn, MatPlot, Seaborn

- Developed Machine Learning models for credit risk prediction and loan default analysis using Logistic Regression, Random Forest, SVM, Decision Tree, and XGBoost on financial datasets.
- Engineered and optimized features using Python, SQL, Pandas, and scikit-learn by applying SMOTE balancing, GridSearchCV, ROC-AUC, Precision-Recall, and F1-Score evaluation techniques to address class imbalance issues.
- Achieved 91% model accuracy, with stronger prediction performance for non-risk customers due to dataset imbalance, while analyzing limitations in identifying high-risk customers.

EDUCATION:

Master of Technology in BioTechnology | 2021-2023 |

National Institute of Technology Warangal, Telangana | CGPA: 7.28 |

TECHNICAL SKILLS:

- **Programming Languages:** Python, SQL
- **Python Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Data Visualization and Analysis:** Power BI, Tableau
- **Database Management:** SQL, NoSQL (MongoDB)
- **Machine Learning:** Supervised Learning (Regression, Classification), Unsupervised Learning (Clustering, Dimensionality Reduction), Ensemble Methods, Neural Networks
- **Deep Learning:** Artificial Neural Networks (ANN), Convolutional Neural Networks, LSTM, GRU, Embeddings (NLP)
- **Deployment:** Flask, Django, FastAPI, Streamlit, Render, Hugging Face Spaces, Git, GitHub
- **Time Series Analysis:** ARIMA, Prophet
- **Cloud:** AWS (Amazon Web Services) - EC2

SOFT SKILLS:

- Problem Solving and Analytical Thinking
- Communication and Presentation Skills
- Critical Thinking and Decision Making
- Teamwork and Collaboration

CERTIFICATIONS / TRAINING

- Certified Data Scientist Training – DataMites, Bangalore | July 2025
- Python for Data Analysis & Visualization & Python Bootcamp (Udemy) | November 2025
- AWS Cloud Practitioner Essentials | November 2025
- Certified Data Scientist International Association of Business Analytics (IABAC) | December 2025